L1 SOC ALERT TRIAGE REPORT

Linux Privileged Command Execution Monitoring Using Splunk

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1. OBJECTIVE & EXECUTIVE SUMMARY

During my Week 5 practical training at DG Interns Hub, I investigated privileged command executions on our Ubuntu server (`ubuntu-soc`) using Splunk Enterprise 10.6.1. Monitoring commands executed with administrative rights (via `sudo` or `pkexec`) is essential for SOC analysts to detect potential abuse of privileges, unauthorized system configuration changes, or post-exploitation activity.

> **Investigation Summary & Disposition**
> Using Splunk to query `/var/log/auth.log`, I analyzed **28 total matching events** capturing commands executed with elevated rights.
>
> **Finding:** All recorded events correspond to expected administrative actions (Splunk service startup, shutdown, status checks, ownership fixes via `chown`, and group assignments via `usermod`) performed by user `gunjan` during lab environment setup, along with automated Ubuntu system update locks (`package-system-locked`) triggered by `pkexec`.
>
> **Verdict: CLOSED - BENIGN ADMINISTRATIVE ACTIVITY**.

2. LAB ENVIRONMENT DETAILS

Component	Technical Specification / Parameter
SIEM Tool	Splunk Enterprise 10.6.1
Operating System	Ubuntu Linux (Oracle VirtualBox)
Hostname	`ubuntu-soc`
Log File Path	`/var/log/auth.log`
Splunk Index & Type	Index: `main` | Sourcetype: `linux_secure`
Primary Executing User	`gunjan` (UID: 1000)

3. CORE TECHNICAL CONCEPTS

Privileged Command Execution: Running system utilities with administrative (root) rights. Root access grants full control over the operating system, including reading sensitive files, modifying services, and creating user accounts. **sudo (Superuser Do):** A Linux command that permits allowed users to execute specific commands as root or another user after authenticating with their password. **Linux Authentication Logs:** The log file `/var/log/auth.log` records all security-sensitive operations, including SSH logins, `sudo` invocations, PAM messages, and group modifications. **chown (Change Owner):** A command used to alter file or directory user and group ownership (e.g., `chown -R splunk /opt/splunk` grants the `splunk` user control over its installation directory). **Splunk Service Commands:** Control scripts used to manage the SIEM service daemon (`splunk start`, `splunk stop`, `splunk status`). **pkexec (PolicyKit Executive):** An authorization tool that allows authorized users to execute commands with elevated root rights according to PolicyKit rules (often used by automated GUI updaters like `package-system-locked`).

4. PRACTICAL INVESTIGATION & SPL QUERY BREAKDOWN

To isolate all privileged command execution records in our ingested security logs, I executed the following search in Splunk Web:

```
index=main sourcetype=linux_secure "COMMAND=" | table _time host _raw | sort -_time
```

Query Explanation:

`index=main sourcetype=linux_secure "COMMAND="`: Filters logs for Linux authentication records containing the keyword `COMMAND=`, which is the standard string logged by `sudo` whenever an elevated command is executed. `| table _time host _raw`: Formats the matching events into a clean tabular layout showing event timestamp, host, and raw log payload. `| sort -_time`: Sorts the results in descending chronological order (newest events first).



Figure 1: Initial Splunk search returning 28 matching `COMMAND=` events across 3 pages of results.
Shows raw event stream in Splunk Web under `index=main sourcetype=linux_secure "COMMAND="` on host `ubuntu-soc`.



Figure 2: Formatted table view (Page 1) detailing recent Splunk service and system package lock commands.
Displays table output sorting commands chronologically, including `splunk start`, `splunk status`, `splunk stop`, and `package-system-locked`.

Page 2 and Page 3 Evidence Review

Continuing through the 28 matching records in Splunk, I reviewed earlier commands recorded during the initial lab configuration session:



Figure 3: Formatted table view (Page 2) showing directory ownership changes and service status queries.
Highlights `chown -R splunk /opt/splunk` executed by `gunjan` to set directory ownership permissions prior to service start.



Figure 4: Formatted table view (Page 3) showing routine `package-system-locked` notifier events.
Records automated background execution of `/usr/lib/update-notifier/package-system-locked` via `pkexec`.

5. CHRONOLOGICAL EVENT TIMELINE & FINDINGS

Analyzing the 28 matching log entries across all 3 pages yields a clear chronological sequence of events (timestamps recorded in +05:30 IST):

Timestamp (+05:30)	User / Process	Command String / Raw Log Description (`_raw`)
20:28:03.020	gunjan (sudo)	COMMAND=/opt/splunk/bin/splunk start (Manual service startup)
20:27:02.590	gunjan (pkexec)	COMMAND=/usr/lib/update-notifier/package-system-locked (Automated OS lock check)
19:14:15.278	gunjan (sudo)	COMMAND=/opt/splunk/bin/splunk status (Manual daemon status check)
19:11:23.196	gunjan (sudo)	COMMAND=/opt/splunk/bin/splunk start (Manual service restart)

Timestamp (+05:30)	User / Process	Command String / Raw Log Description (`_raw`)
19:10:46.340	gunjan (sudo)	COMMAND=/opt/splunk/bin/splunk stop (Manual service stop for configuration)
19:09:47.599	gunjan (sudo)	COMMAND=/usr/sbin/usermod -aG adm splunk (Group assignment for log reading)
17:32:01.152	gunjan (sudo)	COMMAND=/usr/bin/chown -R splunk /opt/splunk (Ownership assignment)

6. SECURITY RISK & MITRE ATT&CK ASSESSMENT

Unauthorized execution of privileged commands poses a major security threat because attackers can use root permissions to disable security controls, create hidden backdoor accounts, or deploy persistent malware.

> **MITRE ATT&CK Mapping Evaluation: No Attack Established**
> While privileged command execution is commonly associated with **TA0004 (Privilege Escalation)** and **TA0003 (Persistence)**, this specific investigation revealed only expected, authorized administrative activity during lab setup. Therefore, **no adversary MITRE ATT&CK technique is mapped to this event**.

7. L1 SOC TRIAGE DECISION & SEVERITY ASSESSMENT

Target Host & User:	Host: ubuntu-soc | User: gunjan
Log Source & Index:	/var/log/auth.log | Index: main (Sourcetype: linux_secure)
Matching Events:	28 total log records across 3 search result pages
Assigned Severity:	LOW / INFORMATIONAL
Triage Disposition:	**CLOSED - BENIGN ADMINISTRATIVE ACTIVITY**
Analyst Assessment:	All 28 events represent legitimate administrative commands (Splunk daemon management, directory ownership fixes, group additions) executed by user `gunjan` during lab setup, alongside routine automated package update checks via `pkexec`. No unauthorized commands, remote connections, or suspicious binaries were observed.

8. RECOMMENDED RESPONSE ACTIONS & STUDENT LEARNING OUTCOMES

L1 Recommended Response Actions:

Maintain baseline monitoring for privileged command execution in /var/log/auth.log using Splunk. If unrecognized commands (such as nc, curl to untrusted external IPs, or credential dumping tools) appear under COMMAND=, immediately collect process trees and escalate to Tier 2. Verify that administrative accounts enforce multi-factor authentication and follow least-privilege principles.

Student Learning Outcomes:

Through this practical exercise, I learned how Linux audits elevated commands via sudo and pkexec in /var/log/auth.log, how to construct Splunk SPL queries to isolate command execution strings, how to parse multi-page search results to build an accurate chronological timeline, and how to properly classify benign administrative setup activity without making false assumptions.

References: Splunk Enterprise Search Reference Guide, Ubuntu Linux Security Auditing Manual, NIST SP 800-92 Guide to Computer Security Log Management.